Final product: retryable batch ingestion of municipal air-quality readings

GitHub repository: https://github.com/yerkebttt/municipal-air-quality-batch-ingestion

Start

Install Docker Desktop or Docker Engine with Docker Compose, clone the repository, open a terminal in the project folder and run:

```
docker compose up --build
```

The first run downloads the UCI dataset and can take several minutes. The loader exits after ingestion; MongoDB keeps running and stores the records in the `mongo_data` volume. `docker compose down` stops the system and keeps the data; `docker compose down -v` also removes the volume and resets the database.

Inspect the result

In a second terminal, connect with the educational credentials defined in Compose:

```
docker compose exec mongo mongosh --username root --password
local-only-change-me --authenticationDatabase admin city_air
db.measurements.countDocuments({})
db.measurements.findOne({station: "Aotizhongxin"})
```

The count should match the 420,768 records of the source. One document is one station-hour with `_id`, station, timestamp, source timezone, source and measurements; missing CSV values are omitted. A compound index supports station and time filters. Batch size and retry limits are set in `docker-compose.yml`.

Limitations

The source must be reachable when ingestion starts. The credentials are for local learning only and must be replaced in any real deployment.

Source

UCI Machine Learning Repository. (n.d.). *Beijing Multi-Site Air Quality*.
https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data